

## Sample 1

### Original English:

The diagram shows a plant cell. Calculate its length.

### Machine Swahili:

Mchoro inaonyesha seli ya mmea. Hesabu urefu wake.

### Refined Swahili:

Mchoro unaonyesha seli ya mmea. Kokotoa urefu wake.

### Issue:

The verb agreement and technical verb choice were inaccurate.

### Why better:

The corrected sentence uses proper grammar and scientific wording.

## Sample 2

### Original English:

In this experiment, temperature is the independent variable.

### Machine Swahili:

Katika hii jaribio, joto ni tofauti huru.

### Refined Swahili:

Katika jaribio hili, halijoto ndiyo kigezo huru.

### Issue:

Noun class and terminology were unnatural.

### Why better:

The refined version uses standard scientific terms used in Swahili education.

### Sample 3

#### Original English:

Bacteria grow faster in warm conditions.

#### Machine Swahili:

Bakteria zinakua haraka kwa hali ya joto.

#### Refined Swahili:

Bakteria hukua kwa kasi zaidi katika mazingira ya joto.

#### Issue:

The machine output was understandable but stylistically weak.

#### Why better:

The correction improves flow and scientific tone.

## Sample 4

### Original English:

Find the average speed of the moving object.

### Machine Swahili:

Tafuta kasi ya kiasi cha wastani ya kitu kinachosonga.

### Refined Swahili:

Tafuta wastani wa kasi ya kitu kinachosogea.

### Issue:

There was redundancy and unnatural word order.

### Why better:

The refined sentence is concise and mathematically natural.

## Sample 5

### Original English:

Describe what happens when the pressure increases.

### Machine Swahili:

Eleza nini kutokea wakati shinikizo inaongezeka.

### Refined Swahili:

Eleza kinachotokea shinikizo linapongezeka.

### Issue:

The sentence had broken grammar and agreement errors.

### Why better:

The correction applies proper tense linkage and noun-class agreement.

## Sample 6

### Original English:

A lot of data were collected during the field study.

### Machine Swahili:

Nyingi data ilikusanywa wakati wa utafiti wa shamba.

### Refined Swahili:

Data nyingi zilikusanywa wakati wa utafiti wa uwandani.

### Issue:

Word order and agreement were incorrect.

### Why better:

The refined sentence uses proper structure and academic register.

## Sample 7

### Original English:

Draw a graph showing the relationship between mass and volume.

### Machine Swahili:

Chora grafu inaonyesha uhusiano kati ya uzito na ujazo.

### Refined Swahili:

Chora grafu inayoonesha uhusiano kati ya massa na ujazo.

### Issue:

The machine translation used less precise scientific vocabulary.

### Why better:

The corrected sentence improves term accuracy and fluency.

## Sample 8

### Original English:

Nutrients are absorbed through the small intestine.

### Machine Swahili:

Virutubisho vinafyonzwa kupitia utumbo mdogo.

### Refined Swahili:

Virutubisho hufyonzwa kupitia utumbo mwembamba.

### Issue:

The anatomical term was not the most accurate for this context.

### Why better:

The refinement uses preferred biology terminology in Swahili.



## Sample 9

### Original English:

The results indicate that the hypothesis was correct.

### Machine Swahili:

Matokeo zinaonyesha kwamba nadharia ilikuwa sahihi.

### Refined Swahili:

Matokeo yanaonyesha kwamba dhana tete ilikuwa sahihi.

### Issue:

There was noun-class disagreement and an imprecise term.

### Why better:

The corrected version uses proper agreement and scientific vocabulary.

## Sample 10

Original English:

Heat the solution until it boils.

Machine Swahili:

Pasha suluhisho mpaka chemka.

Refined Swahili:

Pasha suluhisho hilo hadi lichemke.

Issue:

The machine translation dropped grammatical markers.

Why better:

The correction gives a complete and natural lab instruction.

## Sample 11

### Original English:

Climate change affects biodiversity in many regions.

### Machine Swahili:

Mabadiliko ya tabia nchi inaathiri utofauti wa viumbe katika maeneo mengi.

### Refined Swahili:

Mabadiliko ya tabianchi yanaathiri bioanuwai katika maeneo mengi.

### Issue:

The machine output used awkward and inconsistent terminology.

### Why better:

The refined form uses accepted environmental science terms.

## Sample 12

### Original English:

Catalysts increase the rate of reaction without being consumed.

### Machine Swahili:

Vichocheo huongeza kikomo cha mmenyuko bila kutumika.

### Refined Swahili:

Vichocheo huongeza kiwango cha mmenyuko bila yenyewe kutumika.

### Issue:

The key chemistry term was mistranslated.

### Why better:

The corrected sentence preserves scientific meaning accurately.